

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 12)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain **four** properties of water that make it essential for life.

[4 marks]

0 1 . 2

Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

0 1 . 3

Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

10

0 2 . 1

Explain what a carbon footprint should include when calculated for a car.

[3 marks]

.....

.....

.....

.....

.....

.....

0 2 . 2

Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

.....

.....

.....

0 2 . 3

Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

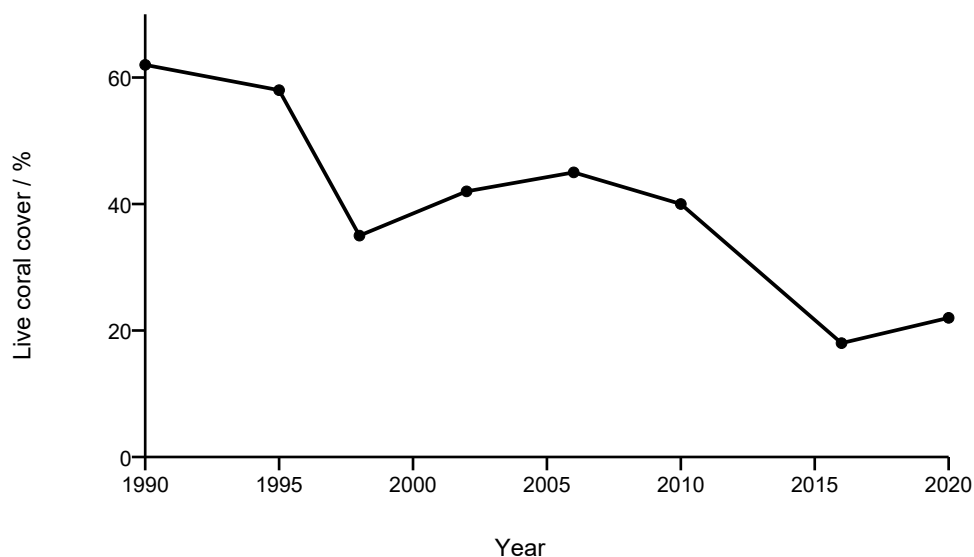
.....

10

03

Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

Figure 1



03

1

Describe the changes in coral cover shown in Figure 1.

[2 marks]

03

2

Explain what coral bleaching is and give **three** causes.

[4 marks]

03 . 3

Explain how the growing resort could threaten the reef other than through bleaching.

[3 marks]

03 . 4

Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

[1 mark]

10

04 . 1

Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

04 . 2

Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

1

2

3

4

0 4 . 3

Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

1

2

3

10

0 5

Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 5 . 1

Complete Table 1 by adding the missing adaptation or application.

[4 marks]

Table 1

Species and adaptation	Design application	
Tubercles (bumps) on the flippers of humpback whales		
	Self-cleaning glass	
Shark skin scales that reduce friction		
	Velcro	

0 5 . 2

Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

0 5 . 3

Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

1

2

0 5 . 4

Explain why bird wing bones are a useful model for engineers.

[1 mark]

10

06 . 1

Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis

☐

B Radio tracking

☐

C Lincoln index (mark-release-recapture)

☐

D Leg rings

☐

06 . 2

Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

.....

.....

.....

.....

06 . 3

Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

.....

.....

.....

.....

5

07 . 1

Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 7 . 2 Describe **five** features of a suitable release site.

[5 marks]

1

.....

2

.....

3

.....

4

.....

5

.....

0 7 . 3 Explain the importance of post-release monitoring.

[2 marks]

.....

.....

.....

.....

10

0 8 Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / m ³ ha ⁻¹ yr ⁻¹	4	16
Bird species	38	9
Ground flora species	60	4

0	8	.	1
---	---	---	---

Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

0	8	.	2
---	---	---	---

Explain why commercial plantations are felled before the trees reach full size.

[2 marks]

0	8	.	3
---	---	---	---

Explain why there is little demand today for coppiced or pollarded wood.

[2 marks]

10

09

Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha⁻¹ yr⁻¹	Food energy output / GJ ha⁻¹ yr⁻¹
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

09**. 1**

Calculate the energy ratio (output / input) for each system.

[2 marks]**Answer**

09**. 2**

Use your answers to explain the difference between productivity and efficiency.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

09**. 3**Explain what is meant by an **energy subsidy** and give **three** examples.**[3 marks]**

.....

.....

.....

.....

09 . 4

Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

10

10 . 1

Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

Extra space

10 . **2** State what is meant by a **fed** aquaculture species.

[1 mark]

Do not write outside the box

10

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
---	---	---	---

Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

OR

1 1 . 2

Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

Do not write outside the box

END OF QUESTIONS